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Can increased automation transparency mitigate the effects of time pressure on automation use?☆

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ABSTRACT

A factor that can potentially negatively impact the accuracy of automated decision aid use, and increase perceived workload, is time pressure. Increased automation transparency can increase the accuracy of automation use. We examined the extent to which increased transparency can mitigate the negative effects of time pressure on the accuracy of automation use and perceived workload. Participants completed an uninhabited vehicle (UV) management task where they assigned the best UV to complete missions by either accepting or rejecting automated advice. Participants made a decision after either 25s (low time pressure) or 12s (high time pressure). The accuracy of automation use decreased, and perceived workload increased, when under higher time pressure. Higher transparency benefited the accuracy of automation use and increased perceived trust and usability. However, high transparency did not mitigate the negative impacts of high time pressure, indicating that increased time pressure can influence the processing of highly transparent information.

1. Introduction

Automation is increasingly used in a variety of contexts to assist humans. An example includes decision support systems, which aid humans in executing tasks by providing a definite recommendation, diagnosis, or solution (Mosier and Manzey, 2020; Strickland et al., 2021). Examples of contexts where decision support systems are used include defence and security (e.g., automated advice regarding the deployment of uninhabited vehicles; Calhoun et al., 2018) and aviation (e.g., aircraft separation assurance decision aids; Trapsilawati et al., 2016). Automation in such contexts is often highly reliable, but can potentially provide incorrect advice. As such, operators may accept incorrect advice (automated advice *misuse*) or reject correct advice (automated advice *disuse*) if they do not correctly verify automated advice or understand the rationale behind the advice (Lee and See, 2004; Parasuraman and Riley, 1997).

Misuse and disuse of automated advice can be partly driven by appropriateness of the level of operator trust in automation (Lee and See, 2004). Formation of trust in automation has been associated with how human-human trust forms (Madhavan and Wiegmann, 2007b), but subtle differences exist between how humans trust automated versus human aids (Madhavan and Wiegmann, 2007a; Pearson et al., 2019). Much like trust in humans, trust in automation may be impacted if the rationale underlying automated advice is not understood, subsequently impacting automation reliance. One factor that may impact trust in and reliance on automation is time pressure (Rice and Keller, 2009).

1.1. Automation use and time pressure

In complex environments, operators often have to make critical decisions under varied time pressure. Research suggests that increased time pressure can lead to a reduction in the quality of information entering the human decision-making process (loss of cognitive *focus*; Boag et al., 2019a; 2023; 2019b; Palada et al., 2018). This reduced quality of information processing under increased time pressure may reflect a reduction in the amount of information processed (i.e., satisficing; Simon, 1956), because it changes the depth of processing (Donkin et al., 2014) and increases the use of decision heuristics (cognitive shortcuts; Gigerenzer and Gaissmaier, 2011; Todd and Gigerenzer, 2000). A reduction in quality of information processing

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under time pressure has been associated with poorer performance due to a speed-accuracy trade-off (i.e., setting lower decision thresholds to favour faster but less accurate responding; Boag et al., 2019a; 2019b; Dutilh et al., 2011; Usher et al., 2002).

In the context of human-automation teaming, a possible heuristic used under time pressure may be to increase decision-dependence (reliance) on the automated advice (i.e., automation bias; Mosier et al., 1996; Parasuraman and Manzey, 2010), which can be associated with increased automated advice misuse (actioning incorrect automated advice). Paradoxically then, several studies have found that when operators use a highly reliable automated decision support tool under extreme time pressure (i.e., 2s to respond), performance can increase due to increased automation reliance (Rice et al., 2008, 2010; Rice and Keller, 2009; Rice and Trafimow, 2012). These findings suggest that time pressure may not be detrimental and can even benefit performance under conditions where automation is highly reliable, and assuming humans heavily rely on the automated advice. However, a limitation of these studies is that the extremely high time pressure may have left participants with no option but to follow automated advice.

Other research has examined human-automation teaming under time pressure which allowed participants time to at least potentially verify automated advice if required (Rieger et al., 2021; Rieger & Manzey, 2022a, 2022b). Specifically, these studies examined how time pressure and highly reliable automated support influenced visual search behaviour and performance, using either a luggage screening task (Rieger et al., 2021; Rieger and Manzey, 2022a) or a medical X-ray screening task (Rieger and Manzey, 2022b). Collectively, and in contrast to prior findings by Rice and colleagues, these studies found no positive effects of time pressure on performance with the use of a decision support tool. Specifically, Rieger et al. (2021) and Rieger and Manzey (2022a) found that the negative effect of time pressure on accuracy was not attenuated with the use of a decision support tool. However, Rieger and Manzey (2022a) found that when participants made a decision prior to receiving automated advice the negative impact of time pressure on performance reduced to little or no effect. Rieger and Manzey (2022b) also found that there was no negative effect of time pressure when using a highly reliable decision support tool, but participants searched through less stimuli and responded faster, indicating some form of underling automation bias.

Given these mixed and limited findings, the current study examines the effect of time pressure on human-automation teaming outcomes. We use automation that is somewhat better or equal to human manual ability, rather than more highly reliable automation that clearly exceeds human manual accuracy. We also use a high time pressure manipulation that allows participants the potential to verify automated advice if required.

1.2. Automation transparency and time pressure

Where operators are under time pressure and interacting with automation, information regarding how automation functions, typically referred to as *automation transparency*, is likely important for achieving real-time situation awareness (Endsley, 2023), and may increase calibrated trust in and reliance on automated advice (Dzindolet et al., 2003; Lee and See, 2004). Automation transparency is intended to aid operator understanding of the rationale underlying automated advice and projected outcomes if advice is followed. A guiding model of transparency design is the Situation-Awareness Agent-Based Transparency (SAT; Chen et al., 2014) model, which outlines three levels of transparency: the automation's goals, purpose, and intentions (Level 1), automation's rationale underling advice (Level 2), and automation's projected future outcomes if advice is followed and any uncertainty associated with those outcomes (Level 3).

Studies have examined the effects of transparency on automation use in a variety of task domains (see Bhaskara et al., 2020; Van de Merwe et al., 2022 for reviews). Increased transparency has typically been found to improve the accuracy of automation use (i.e., correctly accepting/rejecting advice), but some studies have found no benefit (Guznov et al., 2020; Roth et al., 2020), or poorer automation use due to a bias towards agreeing with automation (Bhaskara et al., 2021; Bussone et al., 2015). Increased transparency has also been found to eliminate the increased automation disuse associated with low reliability automation (Gegoff et al., 2023). Furthermore, increased transparency has been found to reduce or have no impact on decision time (e.g., Skraaning and Jamieson, 2021) and perceived workload (e.g., Panganiban et al., 2020). Though findings are inconsistent, increased transparency can increase trust in automation and perceived system usability (Bhaskara et al., 2020; Van de Merwe et al., 2022).

In examining the effects of increased transparency, several studies have used uninhabited vehicle (UV) management tasks (e.g., Bhaskara et al., 2021; Gegoff et al., 2023; Loft et al., 2022; Mercado et al., 2016; Stowers et al., 2020; Tatasciore et al., 2023) and have typically found that increased transparency improves the accuracy of automation use, without cost to perceived workload or decision time. The current study will use a UV management task and aims to replicate these prior findings of benefits to accuracy of automation use with increased transparency, without costs to perceived workload.

The current study further seeks to explore whether transparency can mitigate the potential negative effects of time pressure on the accuracy of automation use. Increased transparency could mitigate the anticipated negative effects of time pressure, given that transparency should allow operators to better judge whether the automation is providing correct advice and the extent to which the information underlying the advice needs to be verified. Assuming increasingly transparent information is presented in an accessible and useable manner, the time required to locate relevant information to verify automated advice may be reduced with increased transparency. As such, the negative impact of increased time pressure on automation use may not be as significant when operators are provided higher compared to lower transparency. To our knowledge, no prior studies to date have examined the impact of time pressure as a function of automation transparency.

However relatedly, Tatasciore et al. (2023) examined the benefits of increased transparency on the accuracy of automation use in the absence or presence of non-automated concurrent task demands. The concurrent task demands manipulation is similar to time pressure in that both impact the time available to assess automated advice. Tatasciore et al. found that although increased transparency increased the accuracy of automation use, there was no evidence to suggest that high transparency mitigated against the negative effects of concurrent task demands. Specifically, the accuracy of automation use declined, and decision time and perceived workload increased, by similar magnitude as a function of concurrent task demands for the high compared to low transparency conditions. In the current study then, high transparency may not mitigate against a reduction in the quality of information entering the human decision-making process under high time pressure if individuals cannot efficiently process the transparency information required to verify automated advice.

1.3. Current study

The current study extended prior research by examining the impact of time pressure on the accuracy of automation use under conditions where automation reliability is slightly better than or equal to human manual performance. Furthermore, the current study aimed to replicate prior findings using the same or highly similar UV management tasks (e.g., Gegoff et al., 2023; Tatasciore et al., 2023) that reported increased transparency could benefit the accuracy of automation use. We also measured perceived workload, trust, and usability. Finally, we explored a novel question concerning whether increased transparency can mitigate the potential negative effects of time pressure on the accuracy of automation use.

The UV management task used had ecological relevance because in

Defence settings operators are required to make decisions under varied time pressure (National Academies of Sciences, Engineering and Medicine, 2022). Participants completed a mission task where they selected the optimal UV to complete a mission based on UV capabilities (e.g., fuel consumption), importance weightings of capabilities, and environmental factors that impact capabilities. Automated assistance (the *Recommender*) was provided which assessed each UV on these attributes and advised the optimal UV. The Recommender was 80% reliable, and as such participants were required to either agree with the automated advice and select Plan A, or to instead chose Plan B. This reliability level was chosen to approximately equate to the manual task performance observed by Tatasciore et al. (2023) and Gegoff et al. (2023), in order to ensure automation reliability was similar to expected human manual ability.

A within-subjects design was used with the two factors of automation transparency (low, high) and time pressure (low, high). Low transparency displays provided information about how the Recommender evaluated the importance of each UV capability based on UV capability weightings (broadly equated to SAT Level 1; Chen et al., 2014). High transparency displays further outlined which UV the Recommender considered to be "better" and "poorer" on each capability. Additionally, high transparency displays provided information about *how* the Recommender calculated each UVs capability scores, including which environmental factors it took into consideration and their projected level of impact (broadly equated to SAT Level 1 + 2 + 3; Chen et al., 2014). Participants made a decision after 25s (low time pressure) or 12s (high time pressure). Low and high time pressure durations were chosen based on the decision times observed by Tatasciore et al. (2023) and Gegoff et al. (2023), in order to ensure participants could potentially verify automated advice under high time pressure if required. Participants were unable to respond to a mission before the timer had elapsed, to ensure participants used all of the time available, thereby maximizing the potential strength of our time pressure manipulation. After the timer had elapsed, participants had 3s to make a decision, during which transparency displays were blanked in order to strictly control the time available to make decisions as a function of manipulated time pressure.

Less accurate automation use and increased perceived workload was expected under conditions of high compared to low time pressure. Furthermore, more accurate automation use, without costs to perceived workload was expected with the use of high compared to low transparency. We then examined the extent to which high transparency mitigated the potential negative effects of time pressure on the accuracy of automation use and perceived workload. The impact of transparency on perceived trust and usability have been more mixed in simulated UV management tasks, so we made no predictions.

2. Methods

2.1. Participants

Participants were 64 (43 female, 18 male, 3 non-binary; $M = 22.03$ years) undergraduate students who took part for course credit and received a performance incentive up to AUD$20. This research complied with the American Psychological Association Code of Ethics and was approved by the Human Resource Ethics Office at The University of Western Australia. Informed consent was obtained.

2.2. Uninhabited vehicle management task

The UV management task was presented on a single desktop monitor (Fig. 1). Participants completed 160 trials (missions), with 80 missions completed under low time pressure (25s) and 80 under high time pressure (12s). Participants were required to deploy the optimal UV to complete each mission. Mission statements were used for face validity

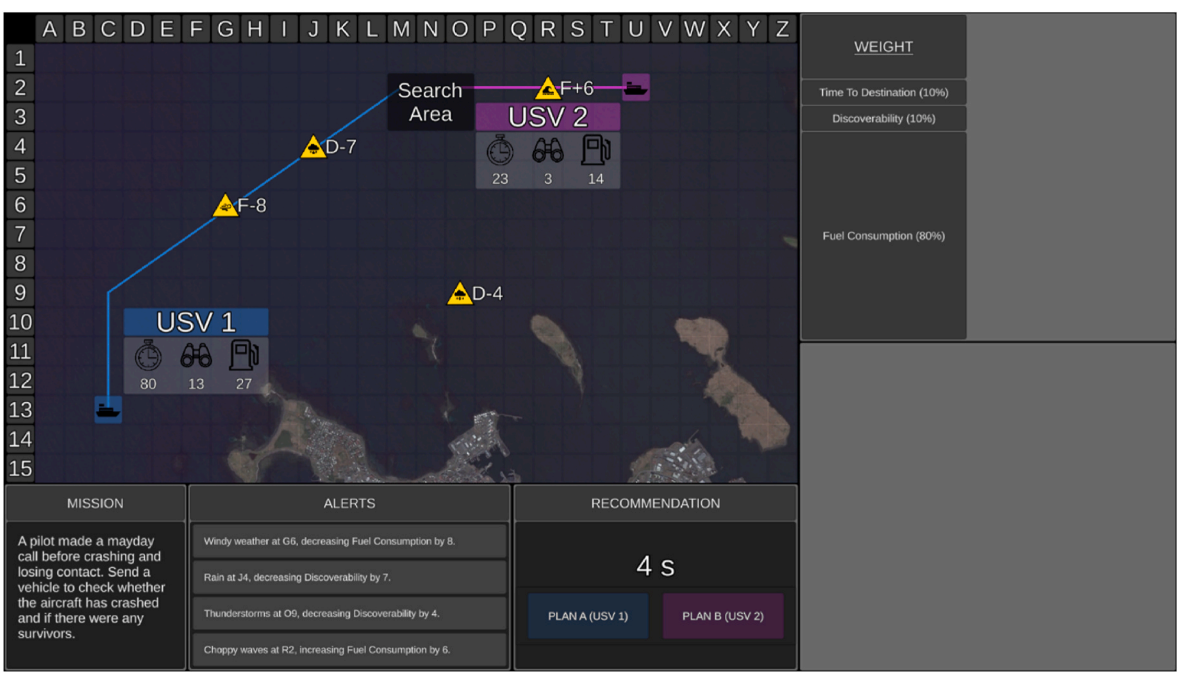

Fig. 1. *The uninhabited vehicle management task with low transparency.*
Note. An example low transparency mission trial with the coastal tactical map. The location of the search area, UV capabilities, the path each UV would take to reach the search area, and environmental hazard symbols are presented on the tactical map. The hazard symbols of relevant environmental factors were placed on the path of UVs. Environmental factors and their impact were also presented in the alerts window. The Recommender's advice can be seen in the Recommendation window. In this example the Recommender has correctly recommended USV 1 as Plan A to complete the mission. Low transparency information is presented in the table (top right) display. The table display presented how the Recommender evaluated the weighting of each capability, with larger rows signifying higher weightings. In this example, fuel consumption has the highest weighting and was thus the only capability that needed to be considered for this mission. The recommendation window also presented the time remaining and two selection buttons.

but were not relevant to UV selection. The tactical map provided an aerial view of rural, coastal, or urban areas. The search area (translucent black box), and two UVs (ground [UGV]; surface [USV]; or aerial [UAV]) were presented on the tactical map. UVs were numbered 1 or 2 and assigned a unique colour (blue or purple), in no particular order. A line was attached to each UV depicting its path to reach the search area.

The optimal UV was determined based on three attributes: UV capabilities (time to destination, discoverability, fuel consumption), capability weightings, and environmental factors that impact capabilities. Time to destination (value presented next to timer symbol in Fig. 1) represented how long the UV would take to reach the search area (lower = quicker). Discoverability (binocular symbol) represented how discoverable the UV was by third parties (lower = less discoverable). Fuel consumption (fuel gauge symbol) represented how much fuel the UV would consume to reach the search area (lower = less fuel).

For each mission, UV capabilities had different importance weightings displayed in the weightings window (Fig. 1). Five weighting sets used varied in difficulty. For hard weighting sets, where two capabilities had equal weightings (e.g., 40%, 40%, 20%), participants were required to consider which UV scored lower on two capabilities. For easy weighting sets (e.g., 60%, 20%, 20%) participants were required to consider which UV scored lower on the highest weighted capability.

Four environmental factors were presented on the tactical map during each mission as yellow hazard symbols, which could be relevant or irrelevant to UV capabilities. If relevant, hazard symbols were presented on the path of a UV (Fig. 1). When irrelevant, factors were not presented on a path of a UV and could be ignored. Hazard symbols represented the type of factor (e.g., thunderstorms), capability impacted (T = time to destination, D = discoverability, F = fuel consumption), direction of impact (+ = positive, - = negative), and level of impact. Four factors and their impact were also presented as messages in the alerts window. During each mission there were one to three relevant factors.

The Recommender advised the optimal UV as Plan A, and the other as Plan B (Recommendation window; Fig. 1), after considering the three UV capabilities. Participants were required to either agree with the Recommender and choose Plan A, or to choose Plan B. Plan A and B buttons were activated after the 25s (low time pressure) or 12s (high time pressure) timer had elapsed. Once the timer had elapsed, the recommendation window flashed to notify participants it was activated, and participants had 3s to make a decision (whereby transparency information was blanked). Feedback was provided after each trial regarding whether the recommendation and participants decision were correct.

Transparency information was presented in the table and graph displays. For low transparency, the table showed how the Recommender assessed the importance of each UV capability, with table rows larger for higher weighted capabilities (Fig. 1). Additionally, for high transparency, the table showed which UV the Recommender considered to be "better" and "poorer" on each capability (Fig. 2).

The additional graph display was only presented with high transparency and included 3 bar graphs (one for each UV capability; Fig. 2). The bar graphs presented how the Recommender calculated the score for each capability, including which factors it took into consideration. Specifically, when an environmental factor was impacting a capability, the hazard symbol was placed on top of the relevant bar on the graph, along with the value added or subtracted from the original score due to the factor. The Recommender's calculated score was then presented in the bar. Shorter bars indicated better capability as lower scores were better. The UV that the Recommender considered to be better on each capability was outlined green. For each graph, Plan A was presented on the left-hand side and Plan B on the right-hand side.

The Recommender was 80% reliable, thereby advising the incorrect UV as Plan A on 20% of the trials. Participants were instructed that "The

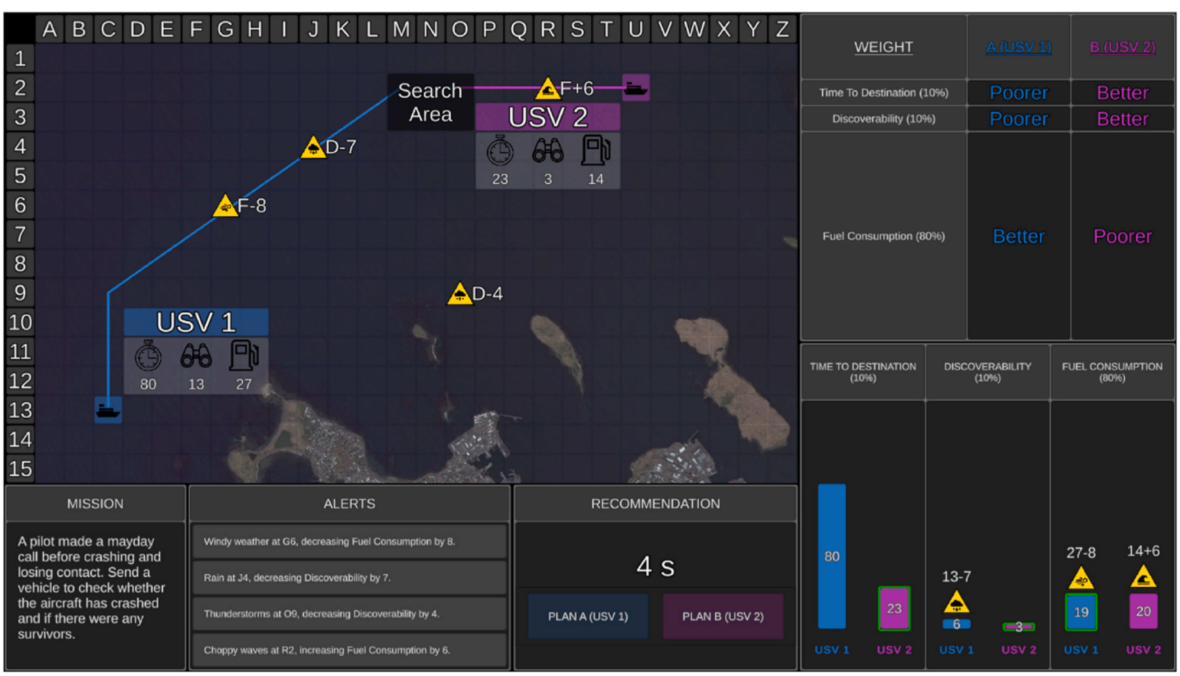

Fig. 2. *The uninhabited vehicle management task with high transparency.*
Note. High transparency information is presented in the table (top right) and graph (bottom right) displays. In both the table and graph displays, Plan A is presented on the left and Plan B on the right. The table display presented how the Recommender evaluated the weighting of each UV capability, with larger rows signifying higher weightings. It also outlined which UV the Recommender considered to be better and poorer on each capability. The graph display presented how the Recommender calculated the score for each capability. When there were relevant environmental factors, the hazard symbol was placed on top of the relevant bar on the graph, along with the value added or subtracted from the original score. The Recommender's calculated scored was presented in the bar. Finally, the UV that the Recommender considered to be better on each capability was outlined green. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

recommender is a highly reliable system, but it is not perfect and as such its recommendation may not be the most optimal plan 100% of the time." This instruction was designed to encourage participants to be generally cautious and thus seek to verify automated advice, while also allowing participants to calibrate their own trust in the automated system. When the Recommender advised the incorrect UV it either: completely missed a relevant factor, miscalculated the impact of a relevant factor, or a combination of both errors across different relevant factors.

For high transparency, if the Recommender missed a relevant environmental factor, the hazard symbol would be missing from the graph display to indicate it was not considered, and the original score would be incorrectly presented instead of the new calculated score. If the Recommender detected but miscalculated the impact of a relevant factor, the hazard symbol would be displayed, but the score that the recommender displayed it had added or subtracted would be incorrect, and the final score would be incorrect. Further, the green border could be placed around the incorrect UVs bar on the graph. Given such errors, in the table display, the UV that the Recommender considered to be "better" and "poorer" on each UV capability could also be incorrect.

On 10% of the reliable trials, the Recommender advised the correct UV as Plan A, but still made one of the three above errors. However, these errors were not significant enough in direction or magnitude to result in the incorrect UV being recommended. These trials were included to discourage participants from simply selecting Plan B as soon as they noticed any type of error made by the Recommender in the graph display (Gegoff et al., 2023; Tatasciore et al., 2023).

The 160 trials were grouped into two 80-trial blocks. In both blocks there was a similar number of relevant environmental factors, capability weighting sets, environmental factor impacts, and type of Recommender errors. The presentation order of blocks was counterbalanced, as well as whether low or high transparency was provided in the first block. The order of unique trials, and time allocated (12s or 25s) to each unique trial, was randomised and yoked such that sets of two participants received the same randomised order of trials and time pressure allocations, but with the presentation order of blocks and transparency differing. The next set of two participants received the same trial order but with the opposite time pressure allocation (e.g., where one participant received 12s for a particular mission, the other participant received 25s for that same mission). Overall, the presentation of blocks, transparency, and assignment of time pressure to specific missions, was counterbalanced across participants.

2.3. Measures

Automated Task Performance. Hit rate was the proportion of trials that the participant selected Plan A when it was correct. Correct rejection rate was the proportion of trials that the participant selected Plan B when the recommended Plan A was incorrect. The Signal Detection parameter d' assessed participants' sensitivity to correctly discriminate Plan A from B. The parameter c assessed participants bias towards agreeing with automation (negative values = greater bias). To adjust for extreme hit or false alarm values (0 or 1), rates of 0 were replaced with 0.5/n, whilst rates of 1 were replaced with (n – 0.5)/n, where n is the number of signal (Plan A correct) or noise (Plan B correct) trials (Macmillan and Kaplan, 1985). Decision times were based on correct decisions.

Workload. Subjective workload was measured using The Air Traffic Workload Input Technique (ATWIT; Stein, 1985) after each mission. Participants had 5s to rate their workload from 1 (very low) to 10 (very high).

System Usability Scale. Perceived usability was measured using an adapted version of the 10-item System Usability Scale (SUS; Brooke, 1996). An example item is "I felt very confident using the Recommender". Items were rated on a 5-point Likert scale from 1 (strongly disagree) to 5 (strongly agree). Some items were reverse-scored and then all items added and multiplied by 2.5 to yield a score from 0 to 100.

Trust Questionnaire. Trust was measured using a six-item questionnaire adapted from Merritt (2011). An example item is "I can depend on the Recommender". Items were rated on a 5-point Likert scale from 1 (strongly disagree) to 5 (strongly agree).

2.4. Procedure

The experiment duration was 2.5hrs. Training began with a 20min audio-visual PowerPoint presentation explaining how to manually complete missions. Participants then completed 20 practice manual trials with 10 completed under low (25s) and 10 under high (12s) time pressure. Participants then watched a PowerPoint presentation specific to the transparency that would be presented in the first block of experimental trials. Participants then completed 80 test trials with the assigned transparency, followed by a 5min break where they completed the trust and SUS questionnaires (in a counterbalanced order). Following this, participants watched a PowerPoint presentation specific to the transparency presented in the second block of experimental trials, after which they completed the remaining 80 test trials. Participants then completed trust and SUS questionnaires.

3. Results

Table 1 presents descriptive statistics for manual training trials as a function of time pressure. Mission accuracy was poorer, $t(62) = 5.04$, $p < .001$, $d = 0.73$, and workload reported as higher, $t(62) = 5.14$, $p < .001$, $d = 0.34$, with high compared to low time pressure.

Table 2 presents descriptive statistics and correlations between dependent variables for the experimental trials, collapsed across transparency and time pressure. As would be expected, a strong positive correlation was found between hit rate and correct rejection rate, and both correlated with sensitivity (d'). There was a strong positive correlation between perceived trust and usability, and a strong negative correlation between workload and usability. Moderate negative correlations were found between workload and hit rate (and sensitivity), and between hit rate and bias (c). Moderate positive correlations were found between usability and hit rate (and sensitivity), as well as between correct rejection rate and bias.

Table 3 presents descriptive statistics for accuracy of automation use and workload as a function of transparency and time pressure. We ran a series of 2 Transparency (low, high) x 2 Time pressure (low, high) within-subjects ANOVAs. Main effects of time pressure were followed up with planned orthogonal contrasts that directly paralleled our hypotheses (Rosenthal and Rosnow, 1985) by comparing low versus high time pressure outcomes at each level of transparency. Descriptives for trust and usability as a function of transparency are presented in Table 4. *T*-tests assessed the impact of transparency on trust and usability. Effect sizes were estimated using partial eta squared (small = 0.01, medium = 0.06, large = 0.14) for *F*-tests and Cohen's d (small = 0.20, medium = 0.50, large = 0.80) for *t*-tests (Cohen, 1992).

3.1. Accuracy of automation use

There was a main effect of time pressure on hit rates, $F(1,63) = 12.07$, $p < .001$, $\eta_p^2 = .16$, with hit rates lower when under high ($M = 0.87$, $SD = 0.10$) compared to low ($M = 0.90$, $SD = 0.09$) time pressure.

Table 1
Descriptive statistics for manual training mission task accuracy and workload as a function of time pressure.

	Low Time Pressure	High Time Pressure
Accuracy	0.86 (.13)	0.75 (.17)
Workload	3.93 (1.62)	4.59 (1.70)

Note. Standard deviations are presented in parentheses.

Table 2
Descriptive statistics and intercorrelations between dependent variables collapsed across transparency and time pressure.

Variable	*M*	SD	1	2	3	4	5	6
1. Hit	.89	.09						
2. Correct Rejection	.76	.15	.65**					
3. Sensitivity *(d')*	2.06	.81	.87**	.91**				
4. Criterion *(c)*	−.30	.18	−.30*	.49**	.10			
5. Workload	4.34	1.67	−.30*	−.27*	−.33**	.05		
6. Trust	2.61	.74	.16	−.06	.03	−.28*	−.27*	
7. Usability	60.14	13.69	.38**	.23	.33**	−.15	−.53**	.63**

Note. Weak, moderate, and strong effect sizes = 0.10, 0.30, and 0.50, respectively (Cohen, 1992). *$p < .05$. **$p < .01$.

Table 3
Descriptive statistics for accuracy of automation use and workload as a function of transparency and time pressure.

	Low Transparency		High Transparency	
	Low Time Pressure	High Time Pressure	Low Time Pressure	High Time Pressure
Hit	.87 (.13)	.85 (.11)	.94 (.08)	.90 (.12)
Correct Rejection	.82 (.19)	.72 (.22)	.80 (.20)	.70 (.20)
Sensitivity (*d'*)	2.16 (1.09)	1.70 (.99)	2.44 (.96)	1.93 (.88)
Criterion (*c*)	−.17 (.25)	−.25 (.28)	−.38 (.30)	−.42 (.34)
Workload	4.23 (1.70)	4.62 (1.77)	4.10 (1.65)	4.44 (1.88)

Note. Standard deviations are presented in parentheses.

Table 4
Descriptive statistics for trust and usability as a function of transparency.

	Low Transparency	High Transparency
Trust	2.42 (.82)	2.79 (.80)
Usability	57.50 (15.40)	62.77 (17.48)

Note. Standard deviations are presented in parentheses.

There was a main effect of transparency, $F(1,63) = 24.16, p < .001, \eta_p^2 = .28$, with hit rates higher with the use of high ($M = 0.92$, $SD = 0.09$) compared to low ($M = 0.86$, $SD = 0.11$) transparency. There was no interaction, $F(1,63) = 1.40$, p = .24, $\eta_p^2 = .02$. With high transparency, hit rates were lower under high compared to low time pressure, $t(63) = 3.66$, $p < .001$, $d = .39$. However, with low transparency there was no difference in hit rates as a function of time pressure, $t(63) = 1.57$, $p = .12$, $d = .17$.

There was a main effect of time pressure on correct rejection rates, $F(1,63) = 29.57$, $p < .001$, $\eta_p^2 = .32$, with correct rejection rates lower when under high ($M = 0.71$ $SD = 0.17$) compared to low ($M = 0.81$, $SD = 0.17$) time pressure. There was no main effect of transparency, $F < 1$, and no interaction, $F < 1$. For both high, $t(63) = 3.72$, $p < .001$, $d = .49$, and low transparency, $t(63) = 4.49$, $p < .001$, $d = .50$, correct rejection rates were lower under high compared to low time pressure.

For sensitivity (d'), there was a main effect of time pressure, $F(1,63) = 48.82$, $p < .001$, $\eta_p^2 = .44$, with participants less sensitive to discriminate Plan A from B when under high ($M = 1.81$, $SD = 0.78$) compared to low ($M = 2.30$, $SD = 0.92$) time pressure. There was also a main effect of transparency, $F(1,63) = 6.03$, $p = .02$, $\eta_p^2 = .09$, with participants more sensitive with high ($M = 2.19$, $SD = 0.84$) compared to low transparency ($M = 1.93$, $SD = 0.97$). There was no interaction, $F < 1$. With both high, $t(63) = 5.43$, $p < .001$, $d = .73$, and low transparency, $t(63) = 5.01$, $p < .001$, $d = .74$, participants were less sensitive at discriminating between Plans A and B under high compared to low time pressure.

For response bias (c), there was a main effect of transparency, $F(1,63) = 29.31$, $p < .001$, $\eta_p^2 = .32$, with participants more biased towards agreeing with the automation with high ($M = -0.40$, $SD = 0.26$) compared to low transparency ($M = -0.21$, $SD = 0.19$). There was no main effect of time pressure, $F(1,63) = 3.51$, $p = .07$, $\eta_p^2 = .05$, and no interaction, $F < 1$.

3.2. Workload

There was a main effect of time pressure on workload, $F(1,63) = 28.48$, $p < .001$, $\eta_p^2 = .31$, with perceived workload higher when under high ($M = 4.53$, $SD = 1.76$) compared to low ($M = 4.17$, $SD = 1.60$) time pressure. There was no main effect of transparency, $F(1,63) = 1.99$, $p = .16$, $\eta_p^2 = .03$, and no interaction, $F < 1$. With both high, $t(63) = 3.67$, $p < .001$, $d = .20$, and low transparency, $t(63) = 5.56$, $p < .001$, $d = .22$, perceived workload was higher under high compared to low time pressure.

3.3. Trust and usability

There was a significant difference in trust and usability ratings, with trust, $t(63) = 4.31$, $p < .001$, $d = .45$, and usability, $t(63) = 2.30$, $p = .03$, $d = .32$, higher with high compared to low transparency.

4. Discussion

We examined the impact of time pressure on the accuracy of automation use under conditions where automation was similar or equal to manual ability and where participants potentially had the time required to verify automated advice. Participants made a decision after 25s (low time pressure) or 12s (high time pressure). We further aimed to replicate prior findings of the benefits of increased transparency on the accuracy of automation use. Finally, we explored whether increased transparency could mitigate the potential negative effects of increased time pressure on the accuracy of automation use and perceived workload. Low transparency provided information about the Recommender's evaluation of the importance of capability weightings. High transparency further outlined which UV the Recommender considered to be better and poorer on each capability, as well as *how* the Recommender calculated each UV's capability scores. Findings are summarised in Table 5.

4.1. Impact of time pressure on automation use

High time pressure resulted in less accurate automation use (sensitivity to correctly discriminate Plan A from B) and increased perceived workload. These findings are inconsistent with prior work that found positive effects of time pressure on automation use (Rice et al., 2008, 2010; Rice and Keller, 2009; Rice and Trafimow, 2012), or no negative effects of time pressure on automation use (Rieger and Manzey, 2022b). A potential reason for these differences is that in the current study reliability was similar to human manual ability, rather than automation being far more highly reliable. Moreover, it is doubtful that participants could have verified automation in the <2s provided in studies conducted by Rice and colleagues.

The current findings are more in line with prior work that has demonstrated that the negative effects of time pressure can remain with

Table 5
Summary of the impact of time and transparency on each outcome measure.

Measure	Impact of Time Pressure	Impact of Transparency	Transparency and Time Pressure
Sensitivity (d')	Poorer sensitivity under high compared to low time pressure	Better sensitivity with high compared to low transparency	No interaction. Planned comparisons indicated that with both low and high transparency, sensitivity was poorer under high time pressure.
Criterion (c)	No difference in bias.	More biased with high compared to low transparency	No interaction. Follow up tests were not planned.
Workload	Increased workload under high compared to low time pressure	No difference in workload.	No interaction. Planned comparisons indicated that with both low and high transparency, workload was higher under high time pressure.
Perceived Trust and Usability	Not applicable.	Increased trust and usability with high compared to low transparency	Not applicable.

the use of a decision support tool (Rieger et al., 2021; Rieger and Manzey, 2022a). A key difference between the current study findings and the prior work of Rieger and colleagues is that in our design participants were unable to further speed-up their response under high time pressure as they were required to wait 12s before making a response (although the same was true for low time pressure conditions who were forced to wait 25s). Despite these caps on decision time, automation use was still poorer when under high compared to low time pressure. Overall, the current findings suggest that in a UV management task context where automation reliability is approximately equal to human manual ability, the negative effects of high time pressure on automation use and perceived workload will likely be observed, and are in line with prior research indicating that increased time pressure can lead to a reduction in the quality of information entering the human decision-making process (Boag et al., 2019a; 2023; 2019b; Palada et al., 2018). It is potentially the case that under high time pressure participants changed their depth of information processing (e.g., Donkin et al., 2014) or increased the use of decision heuristics (Gigerenzer and Gaissmaier, 2011).

4.2. Automation transparency and time pressure

Despite a bias towards agreeing with automated advice, high transparency resulted in more accurate automation use (sensitivity), without costs to perceived workload, compared to low transparency. High transparency use also resulted in higher perceived trust and usability. These findings of increased benefits without cost are generally consistent with prior studies examining the impact of increased transparency in UV management tasks (e.g., Mercado et al., 2016; Gegoff et al., 2023; Stowers et al., 2020; Tatasciore et al., 2023), as well as in other task domains (Van de Merwe et al., 2022). Contrary to several studies using UV management tasks (e.g., Bhaskara et al., 2021; Tatasciore et al., 2023), we found that higher transparency led to increased perceived trust and usability. This may be due to the use of the within-subject transparency manipulation in the current study, that allowed participants to directly compare their experience using low and high transparency automation.

Although high transparency led to benefits, without any associated costs, compared to low transparency, high transparency did not mitigate the observed negative effects of high time pressure on automation use and perceived workload. We expected that high transparency could potentially mitigate these negative effects of time pressure, given that transparency should allow operators to more efficiently judge whether the automation is providing correct advice, and thus the extent to which the information underlying the advice needs to be reviewed. This was clearly not the case, and this finding is in line with the Tatasciore et al. (2023) outcomes that high transparency did not mitigate against the negative effects of concurrent non-automated task demands on automation use. However, based solely on the current data alone we cannot determine the *extent to which* (Boag et al., 2019a; 2019b; Dutilh et al., 2011; Usher et al., 2002) individuals monitored/verified highly transparent information when under high time pressure. Thus, future research is required to objectively examine how operators allocate attention to increasingly transparent information as a function of varied time pressure, such as by using fixation and dwell time eye-tracking metrics or an information masking/uncovering methodology. It would also be diagnostic for future research to use formal methods to assess participant situation awareness of UV capabilities, their weightings, and the impact of environment factors as a function of transparency and time pressure using methods such as the Situational Awareness Global Assessment Technique (Chen et al., 2014; Endsley, 1995), as this would provide an indication of which transparency information has been selectively attended.

Taken together, these findings have important practical implications as although high transparency led to benefits without costs compared to low transparency, the accuracy of highly transparent automation use was not immune to high time pressure. To the extent that designers are aware of what specific high transparency information content is most important for operators, a potential design solution is to cue operators to specific high transparency information as necessary, for example as a function of variation in the expected reliability or the expressed trial-by-trial confidence of automation (Gegoff et al., 2023; also see Strickland et al., 2023). In this case, when operators have limited time available to assess high transparency information (i.e., high time pressure or concurrent task demands), cueing attention to critical information could potentially assist operators to efficiently process necessary information before making a decision (adaptive automation; Greenlee et al., 2022).

4.3. Limitations and conclusion

The UV task used in the current study is broadly representative of work contexts where operators are provided automated advice to assist decision making. However, a limitation was the use of novice participants who undoubtably differ from experts in terms of their cognitive skills and motivation (Rieth and Hagemann, 2022). Although it can be challenging to gather enough experts for required statistical power, future work should examine these same research questions with expert UV operators.

In conclusion, we demonstrated that when automation reliability is similar to human manual performance, the negative effects of time pressure can be observed. Furthermore, high compared to low transparency led to increased accuracy of automation use, perceived trust and usability, without cost to workload. However, high transparency did not mitigate the negative effects of time pressure on accuracy of automation use and perceived workload.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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